



TECHNICAL SUPPORT BULLETIN

Tech. Support Bulletin No.
(NBSB-0005)

Deflection and Friction Test for Ver5 Nanoscope Software

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Products: NS3-NS4a MM
and Dim Systems

Revision	Description	Date
A	Initial Release – John Tedesco	05/02/2007
B	False Engage requirement added – Igor Titushkin	01/07/2016

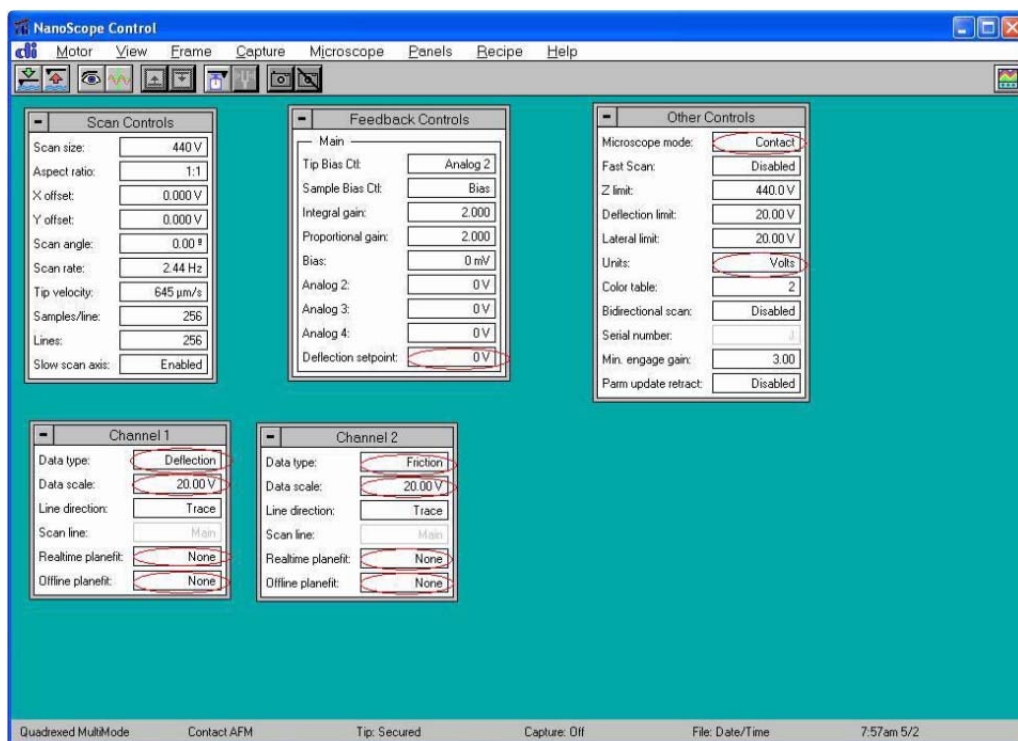
Subject: Deflection and Friction Test for v5 S/W

System(s) Affected: NS3-NS4a MM and Dim Systems with Version 5 software

Action Category: As Required

Summary: To test the main A/D functions on the Z board and the AUX board of the controller

1. Startup Nanoscope software as well as the controller normally. (Note: This test is for version 5 software only).
2. Find your tip with the laser, maximize your SUM signal and center your vertical and horizontal deflections.
3. Set up the parameters as shown in the screen shot below. The important ones have been circled in red. (**Make sure the plane fits are set to NONE**)



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4. It will be necessary to False Engage the microscope in order for the A/D converters to start collecting data. To false engage press Ctrl+Alt+F. The system will start scanning even though the tip is above the surface.
5. Go to View>Scope Trace so that the right hand screen shows two scope views, Ch1 labeled Deflection and Ch2 labeled Friction.
6. Rotate the vertical deflection photo detector adjust knob to move the Deflection scope trace (Ch1) up and down. Rotate the horizontal deflection adjust knob to move the friction scope trace up and down. In the contact mode that we are operating in for this test, the A/D in the AUX board is reading in the friction (horizontal deflection) value and digitizing it, and the A/D on the Z board is reading the deflection (vertical deflection) value and digitizing it. Below are a few pictures to help you locate the vertical and horizontal deflection adjust knobs for both the Multimode and Dimension systems.





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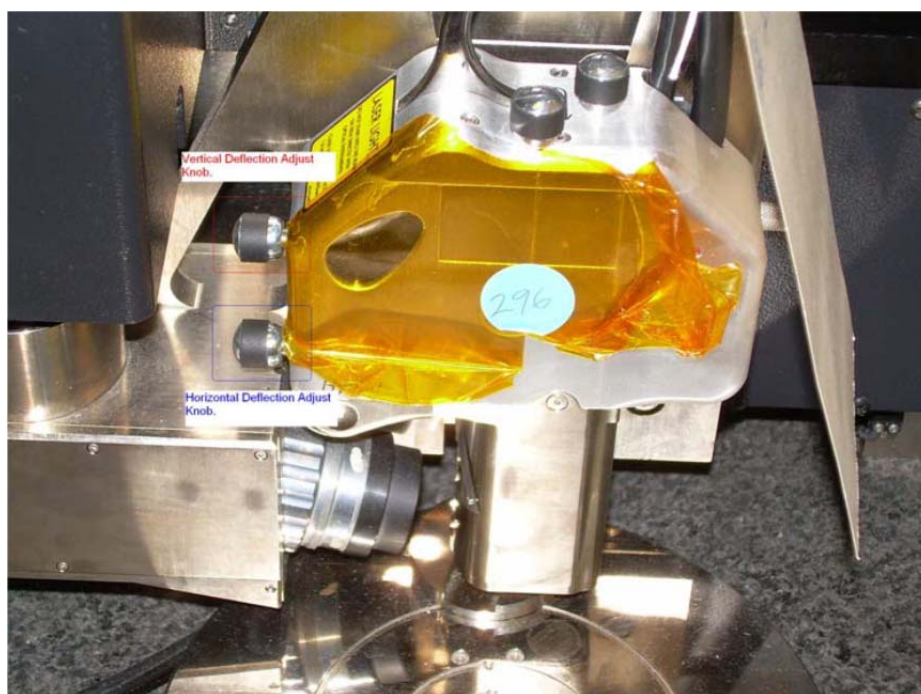
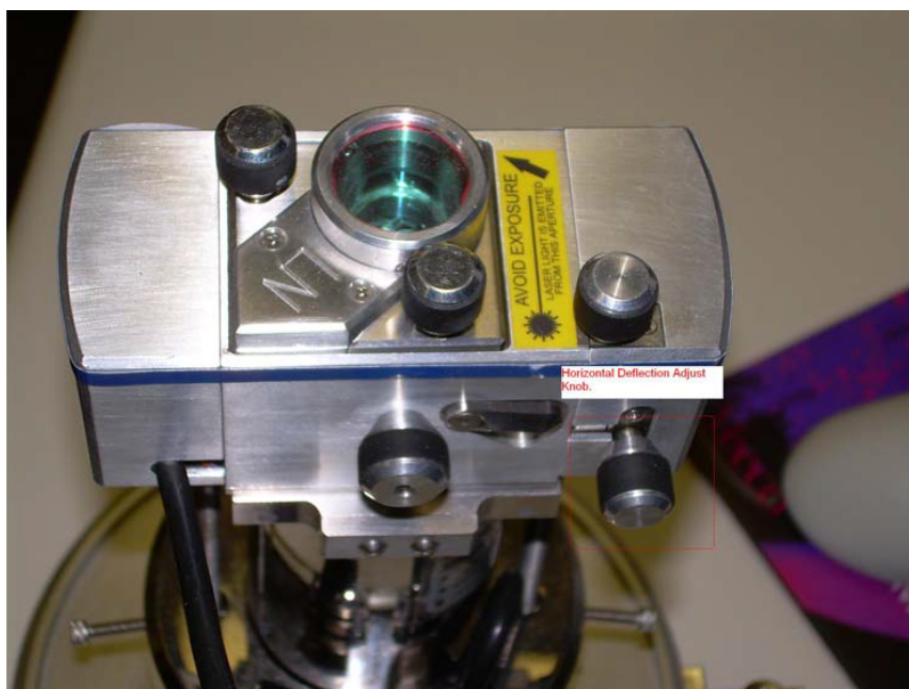
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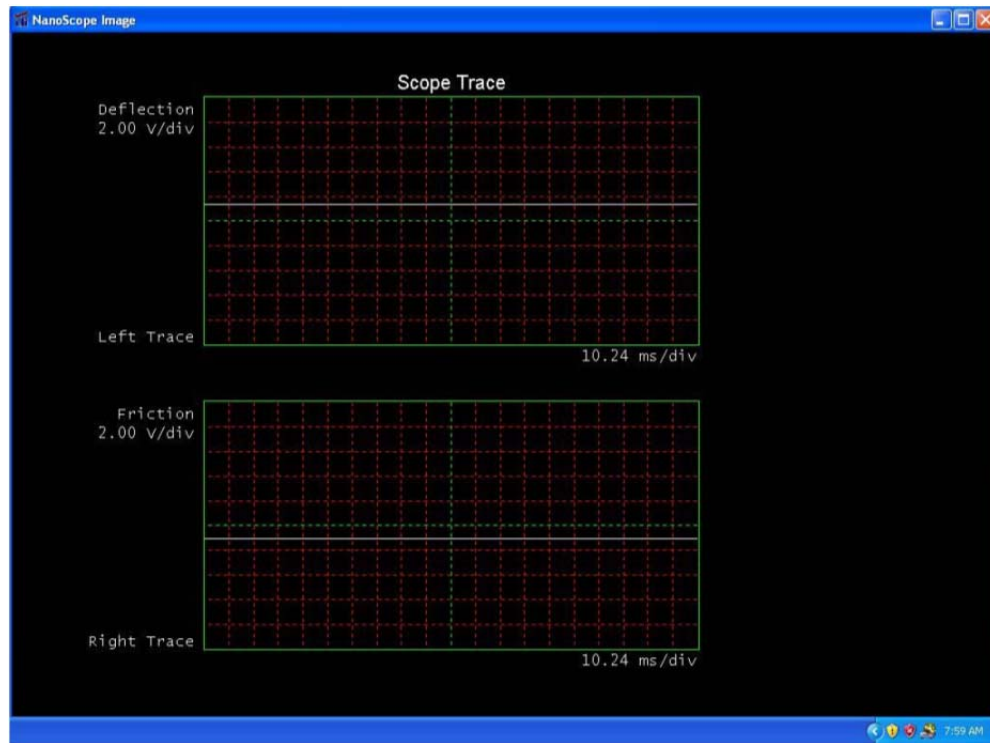


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7. As you move these deflection knobs you should see the scope traces move up and down.
8. Below is an example of the values you would read (on a Dimension system there is no LCD display so this is shown for a Multimode system so that you can see how the values should correspond) if the system were working properly and their corresponding scope traces. You also want to check to make sure that the both A/D converters are reading through the entire range of the 20V possible for both the horizontal and vertical values.



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- Note that both the scope traces correspond with the values read by the Multimode base. Again, with a dimension this is a little harder, you just need to make sure that you can adjust through the entire 20V range for each channel. If that is possible then both A/D converters are working properly.